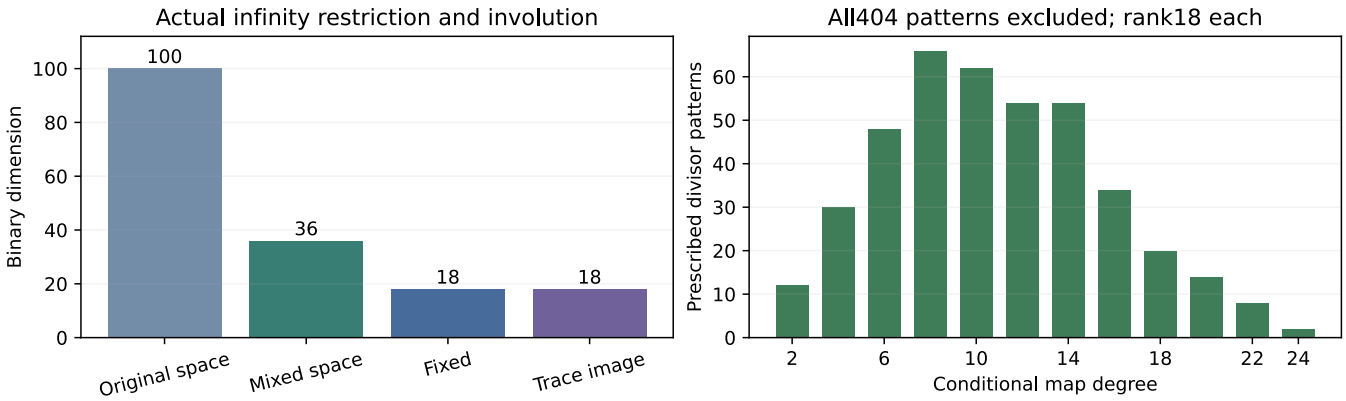
R60: complete infinity-pole space and404 exclusions

The actual complete mixed-pole space has dimension36, with18 fixed functions and trace rank18. A certified trace-one function exists, so both even and odd trace branches are feasible. Every one of the404 prescribed infinity principal-divisor spaces has zero kernel. Each has an exact left inverse, with all source rows reconstructed in a fresh certificate check.



Counts from mixed60-02 and cohort60-01; certificates checked by check60-01. The histogram aggregates404 recorded degree rows. Exact dimensions and exhaustive patterns; no statistical or timing comparison is represented.

Complete pole restriction

Four actual degree2 source points R07,R01,R17,R11 carry original caps 56,8,14,2 in the complete100-dimensional undivided R58 space. For ordinary maps x,t=y/x in that space, signed ramification e_i balances to zero and has minor caps4,7,1. Hence absolute e0 is at most12. Both x and t obey sharper caps24,8,14,2. The first point's orders −56 through −25 give64 binary constraints. Their rank is64, leaving dimension36.

The evaluator uses actual descended quadratic products and all100 source functions, with all four uniquely selected normalized charts, actual bounds and precision checked. Full certificates establish C K=0, R K=I36 and K R+H C=I100. The actual involution restricts and squares to I36. Its N=sigma+I has fixed dimension18 and trace rank18, with complete fixed kernel splitting. A retained particular t has N t=1. Fixed local profile imaginary coefficients vanish in all four charts. The regular514-row source evaluation has a checked left inverse.

Exhaustive ordinary divisor patterns

All9 times15 times3 minor triples exceptzero give404 balanced patterns. On the fixed18-space each case imposes96 binary coefficient rows. Every matrix has rank18 and an explicit H C=I18. The global-regular positive control has rank17 and kernel dimension1 containing the actual constant1. Map degrees run up to24; degree22 cases were included without a coprimality filter. No principal x survives, so there are no scalar or AS branches to run for this cohort. Both trace parities were retained before that gate.

The proper genus61 source, complete R58 space and normalized/DVR chart interpretations are explicit premises. This resolves the stated infinity cohort with both functions in the original bound. Other polar divisors and nonstable F32 finite-support fibers remain available construction spaces.

Exact checks, formal bounds and the cubic enlargement

Attempt	Outcome	Native wall
check60-01	accepted	299 s
cohort60-01	accepted	39 s
infinity-lean60-01	rejected	not recorded
infinity-lean60-02	rejected	not recorded
infinity-lean60-03	accepted	not recorded
mixed60-01	rejected	13 s
mixed60-02	accepted	55 s

Native wall is timestamp-derived, includes this invocation only and is not a performance comparison. Host: migrated Spot workspace, Linux/x86_64; actual source revisions, native binary hashes and UTC are retained per run. Commands: REPLAY_CAS.sh with mixed60-01/02,cohort60-01,check60-01 and the corresponding MIXED60/COHORT60/CHECK60 input and exact marker; REPLAY_LEAN.sh with infinity-lean60-01/02/03. Seed60, one native job/thread, 4GiB virtual and1200seconds each; Lean -j1 -M4096, separate1200second compile/audit and no VM cap. No warmups or statistical sampling.

The first native attempt used an unsupported indexed intvec increment. All four chart bodies failed; its late PASS was rejected and all reported numbers excluded. The repair was frozen before mixed60-02. Two Lean attempts also failed: first an absent Mathlib Omega module, then missing absolute-value notation and recursive enumeration depth. The native table also retains literal t separators because Singular does not interpret the requested tab escape; the renderer parses that recorded format without changing raw bytes. Fresh repairs use core Omega, explicit integer intervals and symbolic finite cardinality identities. No accepted producer was repeated to publish or validate custody.

Fresh certificate invocation

The checker reconstructs all404 constraint matrices as selected rows of the100-function profiles followed by the fixed-basis product. The producer selects rows after forming fixed profiles. All404 reconstructed matrices, all left inverses, mixed and fixed splitting identities, actual odd particular, regular injection and positive control pass. A changed source row fails comparison. This is exact certificate validation, not an independent scientific review or projective-saturation proof.

Three Lean implications

Infinity60.balanced_first_bound proves the sharper absolute e0 bound. balanced_degree_bound proves d=2 sum positive(e) is at most24. nonzero_minor_patterns proves the full404 count by symbolic cardinalities. Lean4.34.1/mathlib d13f23b723b8a846827a245b89c10fc7d3f11612, -j1 -M4096, separate1200second compile/audit limits, no VM cap. Three authored declarations compile and audit; zero failing or trusted native_decide. Source geometry and ramification interpretation remain external premises. See PROOF_MAP.md and the raw audit.

Next complete construction

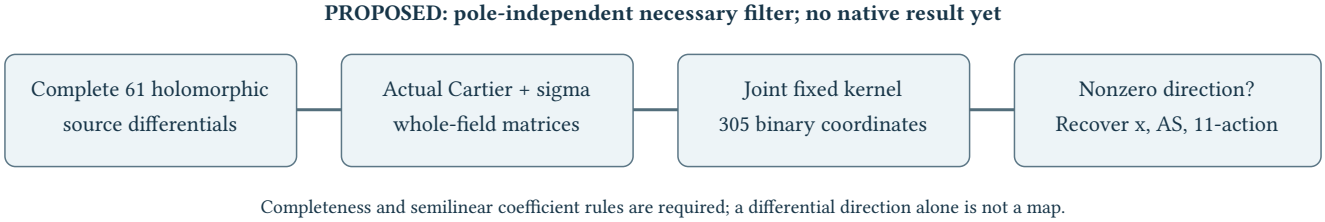
The cubic bound3D permits minor caps6,10,1 and818 balanced nonzero patterns. It forces a common mixed bound34,12,20,2 of degree136. On the retained genus61 source, RR predicts dimension76. First certify all324 actual descended cubic products as a complete degree384 ambient space, using385 regular F4 jets and all20 source charts. Impose all248 binary removed-pole coefficients; require full kernel splitting and actual source function/involution bindings. Dimensions76 and818 are next-stage derived predictions, not R60 measurements. NEXT.md gives the complete acceptance.

Preserve level77 factor/action, full quotient and incidence saturation, proper polarization, node187 coupled divisors, arbitrary reverse classes and ordinary460/367 fibers, direct relations and larger112/991 sources. All original managed owners, native caps and authorizations persist.

Delivery and sources

PR3062 was merged externally at head404f49801. ContinuationPR3200 uses the original stacked base without merge or automatic-merge changes. R59 source science is sealed; its fresh archive download checks all91 files and repacking. R60 records, report and custody follow separately. Sources: R27 normalized charts, R56 divisor geometry, R58 full100-space, R59NEXT, official Singular helper documentation and protocol donor hashes. Read CM.md, results.json, export.json, PROOF_MAP.md, SESSION.md and NEXT.md.

Additional lead: a pole-independent differential filter



Completeness and semilinear coefficient rules are required; a differential direction alone is not a map.

PROPOSED, not measured: for a separable ordinary target map, the pullback of dx/x is a nonzero holomorphic logarithmic differential. Cartier fixes it; sigma(x)=x also makes it sigma-fixed. Compute the complete joint fixed space on all61 source differentials before enumerating every pole bound. A zero joint space would exclude this separable compatible-map class at all bounds; a nonzero space constrains rational-function reconstruction.

The source still needs the complete divisor of a separating du, every exceptional branch, an actual61-form basis and complete whole-field square decompositions f=g squared+u h squared. Construct the true Cartier and sigma matrices; expand inverse squaring on F32 coefficients, c maps to c^16, into305 binary coordinates. Solve the complete F2 joint kernel. Do not treat this semilinear fixed set as an F32-linear space.

A differential direction alone does not supply x, an elliptic map or order11 action. Recover d(x)=x omega modulo squares, prescribed divisors, the AS relation, all5 conjugates and the intended factor/action separately. Inseparable maps require justified Frobenius removal before this gate. The cubic enlargement and other original leads remain on the worklist.

Primary source checked: Friedlander, Garton, Malmskog, Pries and Weir, The a-numbers of Jacobians of Suzuki curves, section3.3, gives the standard Cartier rules and square-decomposition formula. Its Suzuki-specific basis and matrices are not transferred here. Source: Author preprint, section3.3. NEW-LEADS60.md records the full inference, source obligations and checks. R51 already notes that F2 differential matrices cannot certify an integral projector or11-torsion scalar; that action requirement remains separate.

A32-by32 route to the complete differential basis

The existing normal-chart multiplication table supplies T_ij=Trace(e_i e_j). Check its inverse exactly, then use its dual basis for the complementary module Hom over F2[z] of B into F2[z]. Under the normal-source and separability hypotheses, this gives regular differentials on the affine chart after multiplying by dz. Impose all infinity conditions to recover all61 holomorphic forms. Track z=1/u and the differential factor. R27′s complete tame different120 offers an independent divisor check. This is a proposed32-by32 lattice calculation, not a measured trace matrix. The standard module identification is Stacks49.8, lemma49.8.2.